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An agent-based negotiation approach to integrate process planning and scheduling

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This paper presents the development of an agent-based negotiation approach to integrate process planning and scheduling (IPPS) in a job shop kind of flexible manufacturing environment. The agent-based system comprises two types of agents, part agents and machine agents, to represent parts and machines respectively. For each part, all feasible manufacturing processes and routings are recorded as alternative process plans. Similarly, alternative machines for an operation are also considered. With regard to the scheduling requirements and the alternative process plans of a part, the proposed agent-based IPPS system aims to specify the process routing and to assign the manufacturing resources effectively. To establish task allocations, the part and machine agents have to engage in bidding. Bids are evaluated in accordance with a currency function which considers an agent's multi-objectives and IPPS parameters. A negotiation protocol is developed for negotiations between the part agents and the machine agents. The protocol is modified from the contract net protocol to cater for the multiple-task and many-to-many negotiations in this paper. An agent-based framework is established to simulate the proposed IPPS approach. Experiments are conducted to evaluate the performance of the proposed system. The performance measures, including makespan and flowtime, are compared with those of a search technique based on a co-evolutionary algorithm.

Keywords: Integrated process planning and scheduling; Agents; Negotiation; Currency function

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1. Introduction

In manufacturing, both process planning and scheduling functions are responsible for the efficient allocation and utilization of resources. However, the two functions are usually considered separately and sequentially and the process planning decisions are made regardless of the scheduling objectives. If a process plan is prepared off-line without due consideration of the actual shop floor status, it may become infeasible due to changes in the manufacturing environment. The merit of integrated process planning and scheduling (IPPS) is to increase production

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feasibility and optimality by combining both the process planning and scheduling problems (Huang *et al*. 1995). A review of different approaches for solving an IPPS problem can be found in Tan and Khoshnevis (2000). Many researchers, including Nasr and Elsayed (1990), Hutchison *et al*. (1991), Gindy *et al*. (1999), Kim and Egbelu (1999) and Zhang *et al*. (2003), have suggested the integration of process planning and scheduling with alternative process plans during scheduling. This can greatly improve the quality of the schedule and ensure the feasibility of process plans.

Some researchers have focused on the applications of optimization or search techniques in solving the integration problem. Based on mixed integer programming formulations, Nasr and Elsayed (1990) presented two heuristics to determine an efficient schedule for the *n* jobs, *m* machines problem. With the consideration of alternative machines for each operation, they are able to generate job schedules with reducing mean flow times. The restriction is that each job has only a single sequence of operations and a maximum of 10 machines are available in a 10-job problem. Kim and Egbelu (1998) also proposed a mixed integer programming model for an *N*-job, *M*-process plan, *K*-operation IPPS problem, with an objective to minimize the jobs' makespan. However, the approach suffers an exponential increase in the problem complexity for a small increase in problem size. With these exact optimization methods, it is evident that the computational effort grows exponentially as the problem size and degree of scheduling flexibility increase. A typical scheduling problem itself belongs to the class of NP-complete problems; scheduling with alternative process plans is even more difficult due to the enlarged solution space (Khoshnevis and Chen 1990).

Algorithms based on meta-heuristics have been established for solving larger IPPS problems. Weintraub *et al*. (1999) incorporated a tabu search procedure that identified process plans with alternative operations and routings for jobs. The algorithm has been found to be computationally efficient in a manufacturing system involving over 1000 jobs and over 100 machines. More recently, Kim *et al*. (2003) proposed to solve the IPPS problem with a modified evolutionary algorithm called symbiotic evolutionary algorithm (SEA). SEA is based on the idea that parallel searches for different pieces of solution are more efficient than a single search for the entire solution. In their findings, the IPPS problem is decomposed into several sub-problems which are solved by the modified multilateral search algorithm. The proposed algorithm shows an improved search efficiency and diversity over the traditional GA and co-evolutionary algorithms. The performances of the overall system, including jobs' makespan and mean flowtime, are also enhanced in an industrial-size IPPS problem.

In recent years, growing attention has been paid to the development of multi-agent manufacturing control systems. Typical examples are the agent-based system architectures AARIA (Parunak *et al*. 1998), MetaMorph (Maturana *et al*. 1999) and MetaMorph II (Shen *et al*. 2000). These agent-based architectures were developed to support manufacturing activities ranging from design to shop floor control and scheduling. Nevertheless, they do not provide the actual negotiation strategies and protocols governing the interactions and collaborations of agents in specific functions such as shop floor scheduling. The decentralized multi-agent control has been based on the idea that several distributed decision-makers (or agents) can cooperate and interact to obtain globally optimal performances. Negotiation protocols in multi-agent manufacturing systems are mostly based on the contract net

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protocol (CNP) (Smith 1980). Multi-agent systems adopting the CNP have been presented for manufacturing control and scheduling functions in accordance with standard dispatching rules such as first-in-first-out, shortest processing time, critical ratio, or earliest due date (Parunak 1987, Shaw 1987, Baker 1991).

Some agent-based manufacturing systems employ a market-based negotiation mechanism whereby a fictitious currency is introduced to act as a metric for decision making (Lin and Solberg 1992, Krothapalli and Deshmukh 1999, Macchiaroli and Riemma 2002). The advantage of using this currency-based metric is that it is capable of reflecting the part's characteristics and its attractiveness to the machine agents. There are numerous ways to convert the part's attributes into the currency. For instance, Lin and Solberg (1992) used the job's priority to calculate an incentive allocated to a job. Macchiaroli and Riemma (2002) computed the budget as a function of jobs' due dates, tardiness penalty and number of operations. In the currency-based negotiation scheme introduced in Krothapalli and Deshmukh (1999), a currency function is used to evaluate bidding prices according to the due date and the nominal processing time.

A limited number of studies have been conducted to apply agent technology in IPPS. Gu *et al*. (1997) applied a bidding-based approach for machining parts with different features. The system addresses some practical issues for merging the CAPP with shop floor scheduling such as the part's feature representation and operation specification, however, the proposed contract-net based negotiation protocol has a limitation in handling multiple parts concurrently. Usher (2003) employed an agent-based method to dynamically route job orders through production. Although their contract-net based negotiation mechanism coordinates multiple job agents concurrently, each job agent can only announce a single task (or operation) at a time.

As revealed in the literature, most of the conventional IPPS approaches can only achieve the assignment of operations to alternative machines while considering the optimum allocation of resources. Process selection and sequencing still need to be predefined. The integrated approach is mainly to handle the choice of alternative routings in considering the scheduling constraints. This paper presents an agent-based approach for integrating process planning and scheduling. Although there have been a few attempts to applying agent technology to the IPPS problem, they have limitations in the handling of multiple parts or multiple tasks concurrently. The proposed agent-based IPPS system is equipped with a new multi-stage negotiation scheme. It is able to address the real-time process planning and scheduling problems and to cater for the uncertainties and flexibility requirements of modern manufacturing systems.

The problem domain in this paper is specific to the job shop, or similar kind of flexible manufacturing environments, for the production of N parts with M machines. The agent-based system comprises two types of agent, part agents and machine agents, representing parts and machines respectively. Each part agent is provided with the product information including the set of alternative manufacturing processes and routings. The actual process plan for producing a particular product will be established, through negotiation between the part and machine agents, in consideration of the scheduling requirements and availability of manufacturing resources. A negotiation protocol, extended from the CNP, is established to handle multiple-task and many-to-many negotiations. A currency function, which incorporates the important IPPS parameters, processing time and due date, is adopted

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for the preparation and evaluation of bidding prices. A simulation model and experiments are conducted to evaluate the performance of the proposed IPPS approach. The performance measures, including the problem's makespan and the parts' mean flowtime, are compared with those of a search technique based on a co-evolutionary algorithm.

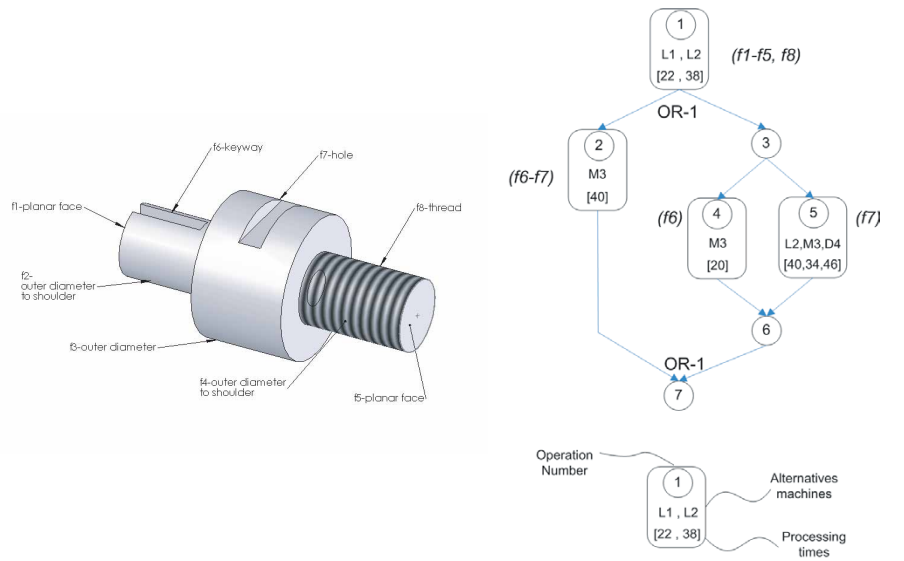
The rest of the paper is organized as follows. Section 2 introduces the problem formulation and overview of the IPPS approach. Section 3 presents the negotiation protocol. A simulation study and its results are shown in section 4. Finally, section 5 provides a summary of the paper.

2. Problem formulation

2.1 *Alternative process plans*

For each part in our IPPS system, all feasible manufacturing processes and routings are recorded as alternative process plans. Similarly, alternative machines for an operation are also recorded. The complete set of feasible production processes and sequences for each part are recorded in an AND/OR graph. The AND/OR graph structure is able to represent the set of possible alternative process plans (Passler *et al*. 1983, Hutchinson and Pflughoeft 1994, Ho and Moodie 1996, Kis 2003). The example part given by Usher and Fernandes (1996) is used here as an example. This simple part (figure 1(a)) consists of eight features (f1–f8) and the corresponding machining processes include turning (f1–f5, f8), milling (f6, f7), and drilling (f7). The list of alternative process sequences are identified in Usher and Fernandes (1996).

There are two types of nodes in our implementation of the AND/OR graph; operation nodes and dummy nodes. An operation node provides the process information of a manufacturing operation. For each operation, the set of alternative



(a) Sample part (from Usher and Fernandes 1996) (b) AND/OR graph

Figure 1. A sample part and its alternative process plans.

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machines and their corresponding processing times are recorded in the operation node. A dummy node does not correspond to any operation and hence it does not provide any process information. The precedence relationship between two connected nodes is represented by the directed edge. Alternative process routes are denoted by an OR-subgraph to indicate that only one of the alternative paths is required to be visited. The network shown in figure 1(b) is the AND/OR graph established to represent the alternative process sequences for the production of this sample part. There is one OR-subgraph where nodes 1 and 7 are the starting and ending nodes respectively. It represents two possible alternative operation sequences for processing features f6 and f7. One is the path with a single process represented by node 2. The other corresponds to the sequence of operations denoted by nodes 3, 4, 5, and 6.

In this paper, an AND/OR graph D is a directed acyclic graph where $D = \{V, A, G\}$. V is the set of nodes, A is a set of directed edges, and G is the set of OR-subgraphs $\{G_1, G_2, \ldots, G_u, \ldots, G_R\}$ corresponding to R sets of OR-subgraphs. Each OR-subgraph G_u has a unique starting operation $S(G_u)$, a unique ending operation $E(G_u)$, and a set of branches $(B_1(G_u), B_2(G_u), \ldots, B_b(G_u))$ where $b > 1$. Each branch corresponds to an alternative path representing a feasible process sequence.

Some operations may be executed with alternative machines. The actual processing time for each operation depends on the choice of machine. To avoid complication, the processing time for an operation with alternative machines is obtained through estimation. The estimated processing time $\hat{t}_j$ of operation j is the average processing time of all its machine alternatives. For example, in figure 1(b), the average processing time of operation 1 $\hat{t}_1$ is 30, which is the average value of L1's and L2's processing times.

Intuitively, the total processing time T for a part with alternative process plans is estimated as follows:

$$T = \sum_{j \notin G_u} \hat{t}_j + \sum_{j \in G_u} \left(\hat{t}_{S(G_u)} + \hat{t}_{E(G_u)} + \frac{\sum_{\forall j \in B_b(G_u)} \hat{t}_j}{b} \right) \quad \forall j \in V \tag{1}$$

The first summation in the equation covers all the operations which do not belong to any OR-subgraph; the second summation incorporates all the OR-subgraphs' estimated processing times. An OR-subgraph's estimated processing time is the sum of the starting operation's processing time $\hat{t}_{S(G_u)}$, the ending operation's processing time $\hat{t}_{E(G_u)}$, and the mean value of processing times of the subordinate operations within the same OR-subgraph. Referring to the OR-subgraph in figure 1(b), for instance, the first branch of the OR-subgraph has only one operation (operation 2) and hence it requires 40 unit time. The second branch consists of two operations (operations 4 and 5) and then the processing time equals $20 + 40 = 60$ unit time. Hence, the mean processing time of these two alternative routes is $(40 + 60)/2 = 50$ unit time. Since the starting operation (operation 1) requires 30 unit time and the ending operation requires 0 unit time, the estimated process plan's processing time T is therefore $30 + 0 + 50 = 80$ unit time.

2.2 *Processing costs*

A part can be produced with alternative process plans and each operation j can be performed on alternative machines. The processing time of performing operation j

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on machine k is defined as t_{jk}. The total production cost (pc_{jk}) for processing operation j with machine k is given by the sum of the fixed cost (fc_{jk}) and variable cost (vc_{jk}). In our model, the variable cost vc_{jk} is determined by the operation's processing time and the part's scheduling priority. The fixed cost fc_{jk} mainly concerns with the overhead and setting up costs of using machine k, which are independent of the processing time or the part schedule. However, the fixed cost can be artificially adjusted to reflect the preference of using a particular machine.

To determine the variable cost, the currency conversion function used by Krothapalli and Deshmukh (1999) is adopted in our IPPS system. The function has been found to be appropriate in similar job shop environments with respect to the effect of due date and processing time. To process a particular operation j with machine k, the variable cost vc_{jk} is expressed as a function of the machine-specific processing time t_{jk} and the shape factor A:

$$vc_{jk} = 2\tan^{-1}\left(e^{t_{jk}A}\right) \tag{2}$$

The shape factor A determines the curvature of the currency function. It is a function of the part's critical ratio cr, as shown below:

$$A = 0.11 - \frac{1 + \tanh(cr - 1)}{20} \tag{3}$$

The critical ratio is one of the commonly used dispatching rules to prioritize parts in a job shop. With a due date d and the processing time of the part T, the critical ratio cr is given by:

$$cr = \frac{d}{T} \tag{4}$$

A lower critical ratio indicates that the part has a tighter due date or longer processing time. A part with a lower critical ratio implies a higher priority than other parts. With the above currency function, an operation from a part with a lower critical ratio (higher priority) or an operation requiring a long processing time will induce a higher variable cost.

2.3 *Solution strategy*

To deal with the production of N parts with M machines in the job shop, the concerned part and machine agents have to negotiate to generate the required process plans in accordance with their respective objectives. In the current implementation, a part is given the objectives to minimize flowtime and to minimize manufacturing cost. Emphasis of the objectives can be adjusted with their respective weightings w_f and w_c where $0 \leq (w_f, w_c) \leq 1$ and $w_f + w_c = 1$. The objectives of the machines are to maximize utilization rate and to maximize revenue, and the respective weightings are w_u and w_r where $0 \leq (w_u, w_r) \leq 1$ and $w_u + w_r = 1$.

To demonstrate the solution approach of the negotiation-based IPPS system, a simple 2-part-4-machine example is used. Figure 2 presents the AND/OR graphs for two parts. Part agent 1 represents the example part shown in figure 1 and part agent 2 represents another part. The part agents are given with their flexible process plans and alternative machines. Four types of manufacturing resources, two lathes,

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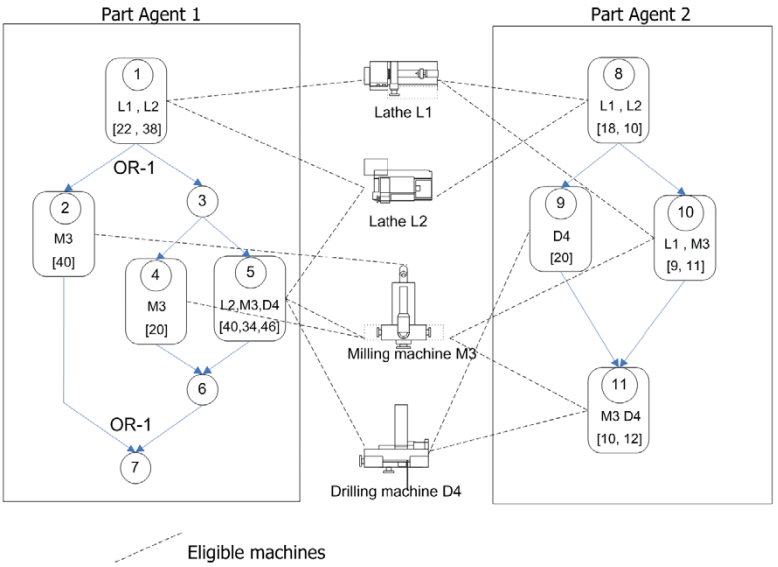


Figure 2. An IPPS problem involving two parts and four machines.

Table 1. Parameters of the illustrative example.

Part	1				2				
Due date (d)	100				100				
Estimated total processing time (T)	80				50				
Critical ratio (cr)	1.25				2.00				
Operation (j)	1	2	4		8		9	10	
Machine (k)	L1	L2	M3	M3	L1	L2	D4	L1	M3
Processing time ($\hat{t}_j$)	22	38	40	20	18	10	20	9	11
Variable cost (vc_{jk})	2.47	2.82	2.84	2.41	1.96	1.79	2.00	1.77	1.81
Fixed cost (fc_{jk})	0	0	0	0	0	0	0	0	0
Production cost (pc_{jk})	2.47	2.82	2.84	2.41	1.96	1.79	2.00	1.77	1.81

one milling machine and one drilling machine, are represented by the corresponding machine agents, L1, L2, M3, and D4, respectively.

It is assumed that all the part agents and machine agents treat their objectives equally by setting the same value of the objective weightings (i.e. $w_f = w_c = w_u = w_r = 0.5$). Let the two parts have the same due date as 100 unit time ($d = 100$). According to the individual operation time elements, the estimated total processing times T of their process plans are 80 and 50 respectively (equation (1)). Their critical ratios cr and costs are then calculated (equations (2), (3) and (4)) and the results are given in table 1.

To begin with, each part agent will invite eligible machine agents to bid for the first available operation. As shown in figure 2, part agent 1 and part agent 2 will select operations 1 and 8 respectively to announce them to the eligible machine agents. If a machine agent is interested in a bid, it will reply with a possible schedule and cost of the operation. The part agent has then to determine the most favourable bid according to its own objectives and hence offer the operation to the winning

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machine agent. After one set of operations has been assigned, the next set of schedulable operations will be announced and the negotiation continues. An IPPS problem is completely solved when all part agents have determined the suitable machines and schedules for all the required operations.

3. Negotiation protocol

A multi-agent negotiation protocol is necessary for effectively coordinating the interactions between the part agents and the machine agents. In this paper, the CNP (Smith 1980) is further extended to support a multi-task, many-to-many negotiation. Figure 3(a) shows the original concept of the CNP which considers one manager announcing one task to multiple contractors. There have been lots of research efforts aiming to extend the original CNP. One common modification is to support bidding between multiple managers and multiple contractors (Lin and Solberg 1992, Macchiaroli and Riemma 2002, Calosso *et al.* 2003). However, they are still limited to a single task announcement in each round of bidding (figure 3(b)). Figure 3(c) shows the modified CNP in this paper. By multi-task negotiation, part agents having several schedulable operations at one time can simultaneously request multiple machine agents for bidding. The motivation is to increase the scheduling efficiency because a part agent will be provided with more bids for making better decisions.

Similar to other agent-based manufacturing applications, the negotiation is part-initiated resource-centred (Lin and Solberg 1992). Part agents and machine agents have to negotiate to establish the shop schedule in accordance with the process plans and operation details recorded in the AND/OR graph. For each round of negotiation, a part agent will search the AND/OR graph to identify the operations which have to be scheduled immediately. The part agent will then announce these unassigned operations to the machine agents. After standard contract net-based negotiation procedures such as task announcement, bid evaluation, and bid confirmation, an operation will either be assigned to a machine agent or it has to be re-announced again if an agreement cannot be obtained with a machine agent. Figure 4 depicts the schematic flow of one round of negotiation between a part agent and a machine agent. Agent interactions and functions of each step and state inside the protocol are listed. A negotiation round begins when a part agent sends a call-for-proposal (CFP) to eligible machine agents to invite them to bid for a particular operation. In figure 4, [*CFP?*], [*BID?*], [*OFFER?*], and [*REPLY?*] denote the states

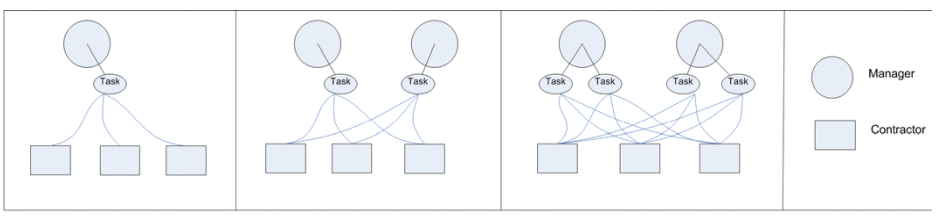


Figure 3. Contract-net protocols.

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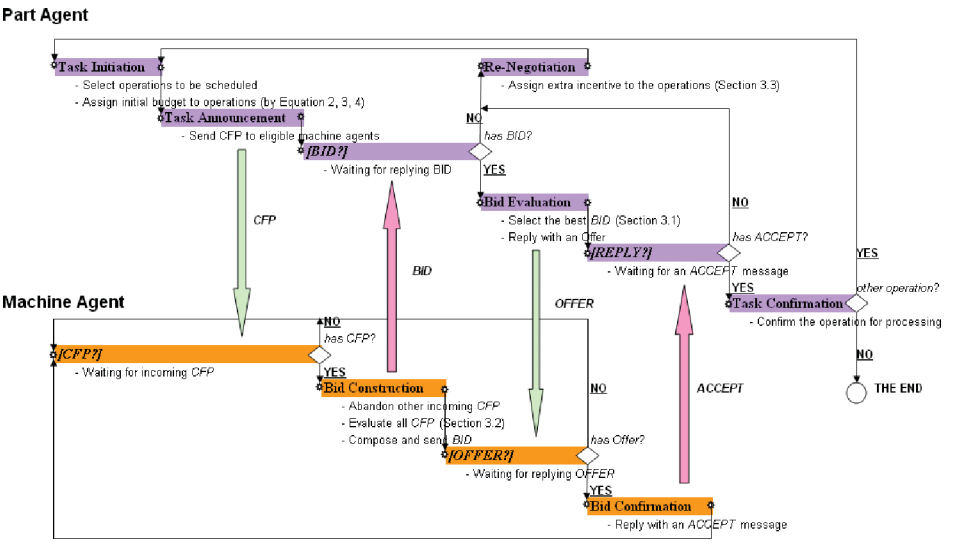


Figure 4. Schematic negotiation flow between the two agents.

of the agents. These states are set with an expiration time to prevent infinite waiting of messages. For instance, a late part agent may not receive any *BID* in the [*BID?*] state because all the machine agents are busy in handling *OFFER* from the early part agents. To avoid infinite waiting, the waiting state of the late part agent will be expired according to the predefined time limit and it allows the part agent to trigger a new round of negotiation. Detailed implementations of bid evaluation by the part agent and bid construction by the machine agent are explained in sections 3.1 and 3.2.

3.1 *Part agent's decision model*

In a heterarchical multi-agent system, an agent has to make decisions according to its own interest while the availability of global information is limited (Duffie and Prabhu 1994). In our work, it is the part agent's responsibility to identify the most favourable *BID* in bid evaluation according to its two objectives (figure 4). The first objective of the part agent (minimize the flowtime) is achieved by selecting the *BID* with a lower value of the operation's finishing time. For an operation j to be processed with machine k, the operation's starting time is X_{jk}, the finishing time FT_{jk} is represented by:

$$FT_{jk} = X_{jk} + t_{jk} \tag{5}$$

The second objective (minimize the manufacturing cost) is measured by the production costs pc_{jk} charged by the machine agent for performing the operation. After all the *BIDs* have been measured, the mean values of the finish times and production cost of all the *BIDs* regarding the same operation are calculated, which are denoted by $\overline{FT}_j$ and $\overline{P}_j$ respectively. A combined *BID* value function is used to identify the most favourable *BID* by considering the two objectives simultaneously. The function incorporates the two objective values, their average values, and the

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corresponding objective weightings. The *BID* with the minimal *BID* value (B_j^*) is selected to be the most favourable *BID* because the objectives of the part agent are to minimize the flowtime and the production cost:

$$B_j^* = \arg\min\left(w_f \frac{FT_{jk} - \overline{FT}_j}{\overline{FT}_j} + w_c \frac{pc_{jk} - \overline{P}_j}{\overline{P}_j}\right) \quad \forall j, k \tag{6}$$

As depicted in equation (6), every FT_{jk} and pc_{jk} is compared with its average value ($\overline{FT}_j$ and $\overline{P}_j$). The aggregated sum of these normalized and weighted differences represents a combined *BID* value.

As seen from the example in figure 2, considering operation 1 of part agent 1, assume both eligible machines (L1 and L2) have replied with *BIDs* and the fixed cost fc_{jk} for each operation is set as zero for simplicity. Using equations (2)–(4), the production costs pc_{jk} of performing operation 1 in L1 and L2 are \$2.47 and \$2.82 respectively (table 1). L1 proposes a finish time of 22 unit time and that of L2 is 38 unit time. Then part agent 1 immediately calculates their average values where $\overline{FT}_1$ is $((22+38)/2)=30$ and $\overline{P}_1$ is $((2.47+2.82)/2)=2.65$. Therefore, the combined bid values from L1 and L2 are $(0.5\times(22-30)/30+0.5\times(2.47-2.65)/2.65)=-0.167$ and $(0.5\times(38-30)/30+0.5\times(2.82-2.65)/2.65)=0.165$ respectively. Part agent 1 will select L1 as the fictitious cost is lower.

3.2 *Machine agent's decision model*

The major decision of a machine agent is to select a *CFP* according to its objectives (maximize the revenue and utilization rate) in bid construction. Similar to the part agent, the machine agent uses a combined *CFP* value to identify the most favourable call-for-proposal *CFP** from the part agents. Assume the revenue R_j earned by the machine agent equals the production cost pc_{jk} specified by the part agent in the *CFP*. Let UR_j denotes the utilization rate of the machine after considering the operation *j*. The mean values of revenue earned and utilization rate of all the *CFP*s are denoted by $\overline{R}$ and $\overline{UR}$ respectively. The machine agent evaluates all the *CFP*s and selects the best call-for-proposal value *CFP** using the following formula:

$$CFP^* = \arg\max\left(w_f \frac{R_j - \overline{R}}{\overline{R}} + w_u \frac{UR_j - \overline{UR}}{\overline{UR}}\right) \quad \forall j \tag{7}$$

where a maximum value is selected because both the objectives of the machine agents are for maximization. After *CFP** is identified, the machine agent has to construct a *BID* specifying the possible starting time X_{jk} and production charge. The possible start time is the earliest starting time of the operation on the machine and the charge equals the production cost pc_{jk} of the operation.

For instance, considering L2 in the first round of negotiation, it receives two *CFP*s from part agent 1 and part agent 2 for performing operations 1 and 8 respectively. Operation 1 requires 38 unit time and will pay \$2.82, whereas operation 8 requires 10 unit time and will pay \$1.79. The average revenue $\overline{R}$ equals $(2.82+1.79)/2=2.31$ and the average utilization rate $\overline{UR}$ is 100% (since it is the first operation performed on the machine). The combined *CFP* value for performing operation 1 then becomes $0.5\times(2.82-2.31)/2.31+0.5\times(100-100)/100=0.11$ and that for operation 8 is $0.5\times(1.79-2.31)/2.31+0.5\times(100-100)/100=-0.11$.

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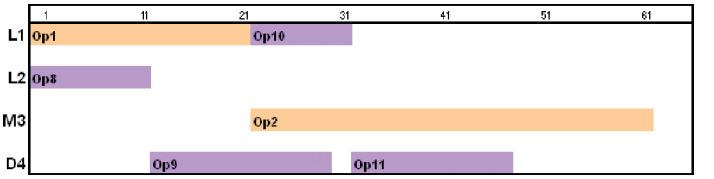


Figure 5. Scheduling result.

Therefore L1 selects operation 1 for bid construction as it has a greater combined value.

3.3 *Re-negotiation*

The proposed negotiation approach includes a *RE-NEGOTIATION* mechanism in any of the following situations: (1) a *CFP* is submitted by the part agent but the target machine agents do not exist or are not in the [*CFP?*] state, (2) the *CFP* is received but is not selected for bid construction by the machine agent, (3) a *BID* is returned but the part agent is not in the [*BID?*] state, (4) a *BID* is chosen after bid evaluation and an *OFFER* is sent by the part agent, but the machine agent finally turns down this *OFFER*, (5) the machine agent accepts the *OFFER* in bid confirmation, but the part agent is not in the [*REPLY?*] state.

In *RE-NEGOTIATION*, the operations (or tasks) are re-announced to the eligible machine agents. An operation is permitted to re-negotiate continuously until a machine has been found. This is to ensure the completeness of the process plan. Before the operation is re-announced, an extra currency is added to the operation's variable cost which may represent an additional cost for negotiation and an incentive to attract the machines. The currency in each round of renegotiation is therefore calculated by,

$$vc_{jk}^{r+1} = vc_{jk}^{r} \times (1+p) \quad \forall r \in \{0,1,\ldots\} \tag{8}$$

where vc_{jk}^{r} is the currency (variable cost) in the rth round of negotiation of the operation j, and p is the increment ratio of the variable cost. The value of p depends on the manufacturer's willingness to pay more on the part in order to release the operation. It can be set at zero if a manufacturer does not agree to pay more for the operation.

With the negotiation-based IPPS approach, the machine selection and scheduling result for the example of figure 2 is generated and depicted in figure 5. Note that for part agent 1, the alternative route with operation 2 is chosen. Therefore operations 4 and 5 are not scheduled.

4. Implementation and experiments

The proposed agent-based IPPS approach is implemented in a Java-based simulation model MAN (multi-agent negotiation). MAN is actually established in the multi-agent development platform JADE (2004). JADE is conformed to the FIPA (Foundation for Intelligent Physical Agent) specifications (FIPA 2004).

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Besides, messages in JADE are encoded in standard ACL (agent communication language). In this paper, all bidding messages (*CFP*, *BID*, *OFFER*, and *ACCEPT*) are compliant with a set of standard performatives of ACL.

4.1 *Experimental setup*

In order to evaluate the performance of the proposed agent-based IPPS, the simulation model MAN is used to conduct a number of experiments. We use the test-bed problems provided in Kim *et al.* (2003) to illustrate the applicability and performance of MAN. Kim *et al.* (2003) generated 18 parts with various combinations of flexibility levels. Each job consists of a minimum of 8 and a maximum of 22 operations. They constructed 24 test-bed problems with the 18 jobs. The number of jobs, the number of operations, and the job composition involved in each problem are listed in table 2. The complete set of data for all 24 problems and 18 parts, including the alternative process plans and the related data, is available in Kim *et al.* (2003) and not repeated here. Since the 18 parts 24 test-bed problems are able to reflect practical manufacturing situations of various complexities, and their approach based on the symbiotic evolutionary algorithm (SEA) is able to address the IPPS problem with good performance (Kim *et al.* 2003), we therefore choose to use the 24 test-bed problems as benchmark cases and to compare the performance of MAN with that of SEA. In the experiments, the manufacturing

Table 2. Test-bed problems.

Problem	Number of jobs	Number of operations	Job number
1	6	79	1 2 3 10 11 12
2	6	100	4 5 6 13 14 15
3	6	121	7 8 9 16 17 18
4	6	95	1 4 7 10 13 16
5	6	96	2 5 8 11 14 17
6	6	109	3 6 9 12 15 18
7	6	99	1 4 8 12 15 17
8	6	96	2 6 7 10 14 18
9	6	105	3 5 9 11 13 16
10	9	132	1 2 3 5 6 10 11 12 15
11	9	168	4 7 8 9 13 14 16 17 18
12	9	146	1 4 5 7 8 10 13 14 16
13	9	154	2 3 6 9 11 12 15 17 18
14	9	151	1 2 4 7 8 12 15 17 18
15	9	149	3 5 6 9 10 11 13 14 16
16	12	179	1 2 3 4 5 6 10 11 12 13 14 15
17	12	221	4 5 6 7 8 9 13 14 15 16 17 18
18	12	191	1 2 4 5 7 9 10 11 13 14 16 17
19	12	205	2 3 5 6 8 9 11 12 14 15 17 18
20	12	195	1 2 4 6 7 8 10 12 14 15 17 18
21	12	201	2 3 5 6 7 9 10 11 13 14 16 18
22	15	256	2 3 4 5 6 8 9 10 11 12 13 14 16 17 18
23	15	256	1 4 5 6 7 8 9 11 12 13 14 15 16 17 18
24	18	300	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

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system is a job shop comprising 15 different machines. During the simulation run and the experiments, MAN was installed in a 2.2 GHz Pentium-based PC.

4.2 *Performance measures*

To evaluate MAN and to compare MAN with SEA, we use the same set of performance measurements in SEA (Kim *et al*. 2003). In every problem instance, the makespan of the problem and the mean flowtime of the parts are measured. The makespan is defined as the amount of time from the start of the first part to the moment the last part is completed. The flowtime of a part is the time required for finishing all the operations of the part. The machines' utilization rate is also measured after every negotiation.

4.3 *Simulation results*

To analyse the effectiveness and robustness of the proposed MAN, several experimental parameters are set which concern different manufacturing environments. Table 3 lists the settings for the experiments. In these six sets of experiments, the parts' due date (d) is set to 1000 and the variable cost's incremental rate (p) is 0.15. Sets 1, 2, 3 and 4 examine the effect of different levels of objective weightings on quality of the schedule. Sets 4, 5 and 6 study the effect of using different negotiation times. For each set of experiments, 10 trials of simulation are carried out and the average result is taken.

4.3.1 Effect of objective weightings. Tables 4–6 show the experimental results of SEA and Sets 1–4 for the makespan, mean flowtime, and mean machine utilization rate respectively. The best method obtained in each problem is identified. The CPU time in table 4 denotes the time required for computation. Note that Sets 1–4 have a similar average CPU time as they are given with the same negotiation time.

In table 4, Set 4 has the smallest average makespan amongst the other methods, and the majority of the best methods are obtained by Set 4 and SEA. On the other hand, regarding the mean part flowtime in table 5, Set 4 reveals a more noticeable advantage over SEA and Sets 1–3. Comparing with Set 2, the superiority of Set 4 for

Table 3. Simulation settings for different scenarios.

Experiment no.	Set 1	Set 2	Set 3	Set 4	Set 5	Set 6
Due date of the parts (d)	1000	1000	1000	1000	1000	1000
Variable cost's incremental rate (p)	0.15	0.15	0.15	0.15	0.15	0.15
Negotiation time (duration for each waiting state)	5 s	5 s	5 s	5 s	1 s	12 s
Part agent's settings						
Weighting of cost (w_c)	0.0	1.0	0.8	0.0	0.0	0.0
Weighting of flowtime (w_f)	1.0	0.0	0.2	1.0	1.0	1.0
Machine agent's settings						
Weighting of revenue (w_r)	1.0	0.0	0.3	0.0	0.0	0.0
Weighting of utilization (w_u)	0.0	1.0	0.7	1.0	1.0	1.0

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Table 4. Effects of objective weightings on makespan (SEA and Set 1–4).

Problem no.	CPU time (s)					Best method
	SEA	Set 1	Set 2	Set 3	Set 4	
1	437.6 (61)	323.0 (125)	348.5 (110)	354.0 (115)	309.0 (121)	Set 4
2	349.7 (69)	417.0 (164)	362.5 (144)	466.0 (164)	323.2 (144)	Set 4
3	355.2 (81)	359.0 (133)	312.5 (115)	366.5 (125)	282.6 (144)	Set 4
4	306.2 (66)	327.0 (134)	392.5 (139)	396.0 (145)	292.8 (124)	Set 4
5	323.7 (64)	288.5 (123)	268.0 (104)	333.0 (114)	282.2 (140)	Set 2
6	443.8 (73)	284.0 (114)	348.5 (124)	310.5 (111)	276.0 (122)	Set 4
7	372.4 (69)	421.0 (144)	375.0 (130)	446.0 (167)	359.8 (128)	Set 4
8	348.3 (67)	325.0 (153)	312.5 (124)	378.0 (126)	250.8 (116)	Set 4
9	434.9 (73)	256.0 (143)	312.5 (115)	362.0 (135)	236.2 (124)	Set 4
10	456.5 (136)	456.5 (145)	401.5 (149)	425.5 (170)	385.8 (149)	Set 4
11	378.9 (166)	446.5 (185)	549.5 (199)	497.0 (179)	463.0 (200)	SEA
12	332.8 (143)	491.0 (167)	518.0 (186)	449.0 (183)	386.8 (159)	SEA
13	469.0 (161)	405.0 (155)	387.0 (146)	381.0 (151)	336.2 (153)	Set 4
14	402.4 (151)	445.0 (165)	502.0 (167)	499.5 (164)	417.2 (171)	SEA
15	445.2 (156)	338.5 (166)	347.5 (138)	373.5 (158)	309.4 (162)	Set 4
16	478.8 (334)	494.0 (203)	492.0 (193)	554.0 (207)	451.0 (207)	Set 4
17	448.9 (435)	518.5 (208)	553.5 (210)	537.0 (214)	495.2 (222)	SEA
18	389.6 (357)	479.0 (187)	518.0 (196)	581.5 (224)	446.4 (196)	SEA
19	508.1 (418)	448.5 (200)	461.5 (199)	489.5 (198)	458.8 (205)	Set 1
20	453.8 (384)	541.0 (200)	515.0 (193)	527.5 (201)	468.8 (210)	SEA
21	483.2 (392)	519.0 (207)	481.5 (192)	546.5 (212)	395.2 (204)	Set 4
22	548.3 (1033)	575.5 (242)	542.5 (282)	651.5 (258)	513.4 (257)	Set 4
23	507.5 (1017)	566.0 (228)	593.0 (267)	671.5 (265)	537.0 (261)	SEA
24	602.2 (1623)	632.5 (274)	712.5 (311)	701.0 (305)	613.8 (315)	SEA
Average	428.2 (314)	431.5 (174)	442.0 (172)	470.7 (179)	387.1 (176)	

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the mean flowtime is attributed to its 'absolute' importance of the part's flowtime weighting as w_f is set to 1 while w_c is 0.

The change of machine weightings also takes effect in the performance of the machines. As depicted in table 3, the weightings of utilization (w_u) in Set 2 and Set 4 are set to 1 and that of Set 1 and Set 3 are set to 0 and 0.7 respectively. As a result, Set 2 and Set 4 have higher overall utilization rates than Set 1 and Set 3 (table 6). With the above results, it is evident that the use of objective weightings and the proposed approach can assist the agents to reach their desirable results.

4.3.2 Effect of negotiation time. As identified in table 3, Sets 4, 5 and 6 are assigned with different negotiation times but all other experimental parameters are identical. The negotiation time means the predefined waiting time of the states including the part agent's [*BID?*] and [*REPLY?*], as well as the machine agent's [*CFP?*] and [*OFFER?*]. Set 4 is given 5 seconds for the waiting states; Set 5 with 1 second; and Set 6 with 12 seconds. Tables 7–9 show the experimental results of SEA and Sets 4–6 for the makespan, mean flowtime, and mean machines' utilization rate respectively.

It is important to see how the negotiation time affects the performances of the proposed approach. In Tables 7–9, all the performances measured in Set 4 and Set 6 are better than those of Set 5. This implies the 1 second of waiting time in Set 5

Table 5. Effects of objective weightings on mean flowtime (SEA and Set 1–4).

Problem no.	SEA	Set 1	Set 2	Set 3	Set 4	Best method
1	318.9	221.8	244.3	240.3	222.7	Set 1
2	287.7	221.7	216.7	233.8	203.6	Set 4
3	304.8	258.8	242.2	255.8	210.4	Set 4
4	251.3	243.8	245.5	255.5	222.1	Set 4
5	280.3	209.1	206.9	220.7	194.4	Set 4
6	384.0	215.3	243.8	236.4	204.4	Set 4
7	314.0	262.2	246.2	277.3	230.9	Set 4
8	295.0	207.8	232.4	235.8	190.6	Set 4
9	298.0	203.5	237.1	255.8	195.3	Set 4
10	349.0	308.1	267.4	298.1	259.3	Set 4
11	312.0	299.6	299.4	320.2	260.7	Set 4
12	279.0	322.7	311.9	305.8	255.8	Set 4
13	387.0	290.7	276.3	277.1	244.9	Set 4
14	346.0	300.7	306.3	313.4	274.4	Set 4
15	316.0	244.9	273.9	279.3	233.4	Set 4
16	359.0	311.4	340.3	357.7	292.4	Set 4
17	364.0	315.7	358.0	338.8	301.6	Set 4
18	322.0	302.3	317.9	337.0	285.4	Set 4
19	406.0	319.1	326.3	333.0	305.9	Set 4
20	372.0	344.2	328.0	349.8	302.0	Set 4
21	365.0	305.0	337.0	380.8	280.1	Set 4
22	417.0	366.2	377.5	386.0	364.9	Set 4
23	404.0	380.5	380.8	403.8	364.3	Set 4
24	452.0	424.1	446.6	431.4	427.3	Set 1
Average	341.0	286.6	294.3	305.1	263.6	

Table 6. Effects of objective weightings on mean machine utilization rate (SEA and Set 1–4).

Problem no.	SEA	Set 1	Set 2	Set 3	Set 4	Best method
1	n/a	60.6%	72.5%	61.1%	71.3%	Set 2
2	n/a	54.2%	69.8%	55.9%	66.2%	Set 2
3	n/a	50.7%	66.3%	61.7%	67.0%	Set 4
4	n/a	46.3%	52.1%	55.3%	55.0%	Set 3
5	n/a	55.5%	71.2%	61.0%	68.7%	Set 2
6	n/a	73.5%	76.5%	69.3%	82.3%	Set 4
7	n/a	46.0%	65.4%	51.9%	58.8%	Set 2
8	n/a	58.3%	62.3%	57.3%	74.2%	Set 4
9	n/a	65.3%	70.4%	60.1%	72.4%	Set 4
10	n/a	63.0%	76.5%	66.0%	78.0%	Set 4
11	n/a	52.9%	62.0%	57.9%	59.6%	Set 2
12	n/a	46.0%	62.3%	56.9%	61.3%	Set 2
13	n/a	66.7%	80.4%	78.2%	84.2%	Set 4
14	n/a	53.7%	64.9%	62.5%	65.4%	Set 4
15	n/a	71.3%	76.8%	69.0%	77.3%	Set 4
16	n/a	67.7%	73.5%	67.7%	75.5%	Set 4
17	n/a	64.1%	66.7%	67.2%	72.3%	Set 4
18	n/a	61.7%	68.3%	60.5%	70.0%	Set 4
19	n/a	74.3%	80.2%	71.1%	78.8%	Set 2
20	n/a	58.2%	72.1%	67.2%	72.2%	Set 4
21	n/a	66.1%	82.0%	64.6%	82.4%	Set 4
22	n/a	72.5%	82.3%	71.3%	76.0%	Set 2
23	n/a	65.7%	73.5%	65.5%	71.9%	Set 2
24	n/a	72.6%	72.2%	75.1%	77.3%	Set 4
Average	n/a	61.1%	70.8%	63.9%	71.6%	

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Table 7. Effects of negotiation time on makespan (SEA and Set 4–6).

	CPU time (s)				
Problem no.	SEA	Set 4	Set 5	Set 6	Best method
1	437.6 (61)	309.0 (121)	293.5 (56)	312.8 (269)	Set 5
2	349.7 (69)	323.2 (144)	352.5 (63)	331.3 (358)	Set 4
3	355.2 (81)	282.6 (144)	336.5 (56)	287.5 (334)	Set 4
4	306.2 (66)	292.8 (124)	433.0 (64)	285.5 (304)	Set 6
5	323.7 (64)	282.2 (140)	292.5 (54)	253.5 (297)	Set 6
6	443.8 (73)	276.0 (122)	292.0 (58)	260.0 (299)	Set 6
7	372.4 (69)	359.8 (128)	368.0 (75)	359.0 (322)	Set 6
8	348.3 (67)	250.8 (116)	315.0 (67)	271.5 (293)	Set 4
9	434.9 (73)	236.2 (124)	295.5 (65)	263.3 (274)	Set 4
10	456.5 (136)	385.8 (149)	403.0 (74)	371.0 (341)	Set 6
11	378.9 (166)	463.0 (200)	466.0 (78)	408.5 (456)	SEA
12	332.8 (143)	386.8 (159)	375.5 (70)	379.3 (390)	SEA
13	469.0 (161)	336.2 (153)	434.0 (76)	343.8 (365)	Set 4
14	402.4 (151)	417.2 (171)	472.0 (82)	429.0 (390)	SEA
15	445.2 (156)	309.4 (162)	320.0 (81)	317.0 (361)	Set 4
16	478.8 (334)	451.0 (207)	526.5 (104)	457.8 (464)	Set 4
17	448.9 (435)	495.2 (222)	488.5 (108)	508.5 (529)	SEA
18	389.6 (357)	446.4 (196)	497.0 (100)	470.3 (445)	SEA
19	508.1 (418)	458.8 (205)	456.0 (105)	420.0 (476)	Set 6
20	453.8 (384)	468.8 (210)	554.5 (103)	482.3 (464)	SEA
21	483.2 (392)	395.2 (204)	528.5 (93)	371.0 (427)	Set 6
22	548.3 (1033)	513.4 (257)	519.0 (95)	544.5 (593)	Set 4
23	507.5 (1017)	537.0 (261)	561.0 (93)	559.8 (543)	SEA
24	602.2 (1623)	613.8 (315)	727.5 (328)	598.8 (620)	Set 6
Average	428.2 (314)	387.1 (176)	429.5 (90)	386.9 (401)	

Table 8. Effects of negotiation time on mean flowtime (SEA and Set 4–6).

Problem no.	SEA	Set 4	Set 5	Set 6	Best method
1	318.9	222.7	241.3	232.0	Set 4
2	287.7	203.6	234.7	198.0	Set 6
3	304.8	210.4	273.5	210.8	Set 4
4	251.3	222.1	269.4	220.8	Set 6
5	280.3	194.4	237.9	188.5	Set 6
6	384.0	204.4	229.8	204.2	Set 6
7	314.0	230.9	257.9	224.2	Set 6
8	295.0	190.6	226.0	197.7	Set 4
9	298.0	195.3	231.0	208.9	Set 4
10	349.0	259.3	284.0	245.4	Set 6
11	312.0	260.7	322.0	240.7	Set 6
12	279.0	255.8	297.8	249.5	Set 6
13	387.0	244.9	333.9	252.7	Set 4
14	346.0	274.4	349.6	278.8	Set 4
15	316.0	233.4	260.5	235.7	Set 4
16	359.0	292.4	339.7	292.0	Set 6
17	364.0	301.6	324.0	300.2	Set 6
18	322.0	285.4	312.2	273.9	Set 6
19	406.0	305.9	355.3	278.6	Set 6
20	372.0	302.0	380.2	307.5	Set 4
21	365.0	280.1	382.0	262.6	Set 6
22	417.0	364.9	390.9	327.5	Set 6
23	404.0	364.3	393.3	329.7	Set 6
24	452.0	427.3	506.6	352.8	Set 6
Average	341.0	263.6	309.7	254.7	

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Table 9. Effects of objective weightings on mean machine utilization rate (SEA and Set 4–6).

Problem no.	SEA	Set 4	Set 5	Set 6	Best method
1	n/a	71.3%	62.5%	68.9%	Set 4
2	n/a	66.2%	62.8%	68.1%	Set 6
3	n/a	67.0%	57.8%	64.9%	Set 4
4	n/a	55.0%	45.3%	54.4%	Set 4
5	n/a	68.7%	56.1%	73.6%	Set 6
6	n/a	82.3%	74.6%	82.1%	Set 4
7	n/a	58.8%	59.9%	61.0%	Set 6
8	n/a	74.2%	66.1%	72.5%	Set 4
9	n/a	72.4%	62.8%	70.6%	Set 4
10	n/a	78.0%	68.2%	78.4%	Set 6
11	n/a	59.6%	55.0%	63.2%	Set 6
12	n/a	61.3%	57.9%	62.1%	Set 6
13	n/a	84.2%	72.1%	82.9%	Set 4
14	n/a	65.4%	54.5%	64.2%	Set 4
15	n/a	77.3%	74.1%	76.1%	Set 4
16	n/a	75.5%	67.0%	71.9%	Set 4
17	n/a	72.3%	69.9%	66.1%	Set 4
18	n/a	70.0%	64.8%	70.2%	Set 6
19	n/a	78.8%	75.9%	82.4%	Set 6
20	n/a	72.2%	67.2%	72.3%	Set 6
21	n/a	82.4%	67.5%	85.4%	Set 6
22	n/a	76.0%	78.1%	77.1%	Set 5
23	n/a	71.9%	71.9%	71.2%	Set 4
24	n/a	77.3%	66.2%	79.5%	Set 6
Average	n/a	71.6%	64.9%	71.6%	

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is inadequate for the multi-agent negotiation. In this 1 second period, the agents may not be able to collect all the *CFP*, *BID*, *OFFER*, or *ACCEPT*. This may lead to insufficient options for them to make better decisions and therefore results in unsatisfactory performance.

4.4 *Comparison of SEA and MAN*

Table 10 compares the results of different performance measures using SEA with those obtained from MAN (based on Set 4).

Regarding the mean flowtime, MAN outperforms SEA with an average improved rate of 22.7% on all problem instances. On the other hand, MAN shows only a marginal advantage of 9.6% over SEA in terms of the makespan. In fact, the performance on makespan is mixed. For example, MAN has an advantage of 45.7% in Problem 9, but a detriment of 22.2% in Problem 11. To explain the contradicting outcomes, it is required to understand that all the agents in MAN are designed to optimize their individual currency function according to their localized information and objectives only. It is understandable for the agents to improve their individual objectives at the expense of the overall system objective. As shown from the findings, the part agents have improved the flowtime (local objective) without consideration of the makespan of the problem (system objective).

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Table 10. Comparison of SEA and MAN (Set 4).

	Makespan			Mean Flowtime			Average Machine Utilization		CPU time (Sec)		
Problem	SEA	MAN (Set 4)	Improved rate %	SEA	MAN (Set 4)	Improved rate %	SEA	MAN (Set 4)	SEA	MAN (Set 4)	Improved rate %
1	437.6	309.0	29.4%	318.9	222.7	30.2%	n/a	71.3%	60.5	121.1	−100.1%
2	349.7	323.2	7.6%	287.7	203.6	29.2%	n/a	66.2%	68.9	143.6	−108.4%
3	355.2	282.6	20.4%	304.8	210.4	31.0%	n/a	67.0%	81.0	144.0	−77.8%
4	306.2	292.8	4.4%	251.3	222.1	11.6%	n/a	55.0%	65.6	123.6	−88.5%
5	323.7	282.2	12.8%	280.3	194.4	30.7%	n/a	68.7%	63.5	140.1	−120.7%
6	443.8	276.0	37.8%	384.0	204.4	46.8%	n/a	82.3%	73.3	122.2	−66.7%
7	372.4	359.8	3.4%	314.0	230.9	26.5%	n/a	58.8%	69.0	127.6	−84.9%
8	348.3	250.8	28.0%	294.0	190.6	35.4%	n/a	74.2%	67.3	116.3	−72.8%
9	434.9	236.2	45.7%	298.0	195.3	34.5%	n/a	72.4%	73.2	123.6	−68.9%
10	456.5	385.8	15.5%	349.0	259.3	25.7%	n/a	78.0%	136.0	149.3	−9.7%
11	378.9	463.0	−22.2%	312.0	260.7	16.5%	n/a	59.6%	165.8	200.5	−20.9%
12	332.8	386.8	−16.2%	279.0	255.8	8.3%	n/a	61.3%	143.4	158.5	−10.5%
13	469.0	336.2	28.3%	387.0	244.9	36.7%	n/a	84.3%	161.2	153.3	4.9%
14	402.4	417.2	−3.7%	346.0	274.4	20.7%	n/a	65.4%	150.8	170.5	−13.1%
15	445.2	309.4	30.5%	316.0	233.4	26.1%	n/a	77.3%	156.0	162.1	−3.9%
16	478.8	451.0	5.8%	359.0	292.4	18.6%	n/a	75.5%	333.6	207.2	37.9%
17	448.9	495.2	−10.3%	364.0	301.6	17.1%	n/a	72.3%	435.2	222.5	48.9%
18	389.6	446.4	−14.6%	322.0	285.4	11.4%	n/a	70.0%	357.0	196.4	45.0%
19	508.1	458.8	9.7%	406.0	305.9	27.7%	n/a	78.8%	417.8	204.7	51.0%
20	453.8	468.8	−3.3%	372.0	302.0	18.8%	n/a	72.2%	384.0	210.1	45.3%
21	483.2	395.2	18.2%	365.0	280.1	23.3%	n/a	82.4%	392.4	203.8	48.1%
22	548.3	513.4	6.4%	417.0	364.9	12.5%	n/a	76.0%	1033.3	256.5	75.2%
23	507.5	537.0	−5.8%	404.0	364.3	9.8%	n/a	71.9%	1016.6	260.6	74.4%
24	602.2	613.8	−1.9%	452.0	427.3	5.5%	n/a	77.3%	1622.7	314.7	80.6%
Average	428.2	387.1	9.6%	341.0	263.6	22.7%	n/a	71.6%	313.7	176.4	−14.0%

The result on machine utilization is not available in SEA. Remarkably, MAN is able to maintain a good performance with an average machine utilization of over 70% for most of the problem sets.

Although it is not meaningful to directly compare the CPU times of MAN and SEA as they were installed in different computing environments (MAN on a 2.2 GHz PC while SEA on a 700 MHz), the CPU times of SEA are also listed for reference as the relationship between the problem complexity and CPU time is of interest. SEA requires a significant surge in computation time for the harder problems such as Problems 22–24 and it is a common drawback for evolutionary algorithms to handle large-scale problems. In contrast, the CPU times of MAN are not so much affected by the problem size. Rather, it is proportional to the negotiation time and it gently increases with the complexity of the problem. As listed in table 7, the CPU times for Set 5 (negotiation time = 1s) are much shorter than those of Sets 4 (negotiation time = 5s) and 6 (negotiation time = 12s) and SEA.

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5. Concluding remarks

As evident in this paper, the multi-agent negotiation approach is an effective approach to solve the integrated process planning and scheduling problem in the job shop kind of flexible manufacturing environments. The negotiation protocol, which incorporates several negotiation states and a re-negotiation mechanism, has enhanced the system's robustness over the previous works as it allows a multi-task negotiation between multiple parts and machines. The decision models of the agents have successfully aided them to make decisions according to their own objectives.

Extensive experiments are conducted in a simulation environment. It is shown that the objective weightings and negotiation time take an obvious effect on the results of the problem. The performances of the proposed agent-based system are compared with those of the search technique based on the symbiotic evolutionary algorithm. The computational experience reveals that, in the pursuit of the local objectives such as the part's flowtime, the proposed approach performs better than the meta-heuristics.

It is clear that much work still needs to be performed in the research of agent-based application in IPPS. The paper has demonstrated that the complex dynamic process planning and scheduling problem can be solved effectively through agent-based negotiation. Yet the proposed approach has an unstable performance from a perspective of the global objectives, such as the makespan of problems. This is mainly because it lacks a central coordination of the agents and it is a limitation of a pure decentralized system without supervisor control (Choi *et al*. 2000). One aspect of our further research is on enhancing the architecture of the current multi-agent system to allow agents to possess a global view for evaluating the global system objectives. One possible approach is to introduce supervisor agents to the system. The supervisor agents may help to evaluate the local decisions in a global manner such that the final decision is more favourable to the system objectives. In another aspect of further research, the negotiation protocol and currency function can be modified to incorporate different manufacturing environments and

practical considerations. For instance, different currency functions can be implemented for part and machine agents to emphasize their respective manufacturing objectives.

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